CURRICULUM VITAE

SHENGBO WANG

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Assistant Professor
Daniel J. Epstein Department of Industrial and Systems Engineering
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EDUCATION

Stanford University — Doctor of Philosophy 2020–2025
Management Science and Engineering
Ph.D. in Management Science and Engineering.
Co-advised by Prof. Peter Glynn and Jose Blanchet.

Cornell University — Bachelor of Science 2017–2020
Operations Research and Information Engineering, College of Engineering
Summa cum laude.
Major: Operations Research (honors program).
Minor: Applied Mathematics.

HONORS AND AWARDS

APS Best Student Paper Award. INFORMS Applied Probability Society, 2025
Meritorious Reviewer Award. Mathematics of Operations Research, 2025
Merrill Presidential Scholar (top 1%). Cornell University, 2020
Byron W. Saunders Award. Cornell ORIE, 2020

PUBLICATIONS

Journal Papers

[1] Shengbo Wang and Nian Si. *Bellman Optimality of Average-Reward Robust Markov Decision Processes with a Constant Gain*. 2025. arXiv: 2509.14203 [math.OC].
Accepted by *Operations Research*.

[2] Julien Grand-Clément, Nian Si, and Shengbo Wang. *Dynamic Programming-Compatible Uncertainty Sets in Robust Markov Decision Processes*. 2024. arXiv: 2411.08435 [math.OC].
Accepted by *Mathematics of Operations Research*.

[3] Shengbo Wang, Nian Si, Jose Blanchet, and Zhengyuan Zhou. "Sample Complexity of Variance-Reduced Distributionally Robust Q-Learning". In: *Journal of Machine Learning Research* 25.341 (2024), pp. 1–77.

Conference Papers

[4] Zhenghao Li, Shengbo Wang, and Nian Si. "Near-Optimal Sample Complexities of Divergence-Based S-Rectangular Distributionally Robust Reinforcement Learning". In: *Proceedings of The 29th International Conference on Artificial Intelligence and Statistics*. Vol. 300. Proceedings of Machine Learning Research. PMLR, 2026.

[5] Zijun Chen, Shengbo Wang, and Nian Si. "Sample Complexity of Distributionally Robust Average-Reward Reinforcement Learning". In: *Advances in Neural Information Processing Systems*. Vol. 38. Curran Associates, Inc., 2025, pp. 85402–85463.

[6] Shengbo Wang, Nian Si, Jose Blanchet, and Zhengyuan Zhou. "Statistical Learning of Distributionally Robust Stochastic Control in Continuous State Spaces". In: *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*. Vol. 258. Proceedings of Machine Learning Research. PMLR, 2025, pp. 2791–2799.

[7] Shengbo Wang, Jose Blanchet, and Peter Glynn. "An Efficient High-dimensional Gradient Estimator for Stochastic Differential Equations". In: *Advances in Neural Information Processing Systems*. Vol. 37. Curran Associates, Inc., 2024, pp. 88045–88090.

[8] Shengbo Wang, Jose Blanchet, and Peter Glynn. "Optimal Sample Complexity for Average Reward Markov Decision Processes". In: *The Twelfth International Conference on Learning Representations*. 2024.

[9] Shengbo Wang, Nian Si, Jose Blanchet, and Zhengyuan Zhou. "A Finite Sample Complexity Bound for Distributionally Robust Q-learning". In: *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*. Vol. 206. Proceedings of Machine Learning Research. PMLR, 2023, pp. 3370–3398.

Preprints

[10] Zijun Chen, Zaiwei Chen, Nian Si, and Shengbo Wang. *Achieving ε^{-2} Dependence for Average-Reward Q-Learning with a New Contraction Principle*. 2026. arXiv: 2601.21301 [cs.LG].

[11] Shengbo Wang. *Q-Learning with Stable Infinite-Dimensional Linear Function Approximation*. 2026. arXiv: 2608.22636 [cs.LG].

[12] Shengbo Wang. *Q-Measure-Learning for Continuous State RL: Efficient Implementation and Convergence*. 2026. arXiv: 2603.03523 [cs.LG].

[13] Shengbo Wang and Jose Blanchet. *Lipschitz Regularity in Wasserstein Robust Stochastic Optimal Control*. 2026. arXiv: 2606.28633 [math.OC].

[14] Shengbo Wang, Jose Blanchet, and Peter Glynn. *Fast Convergence of Policy Regret in Learning Stochastic Optimal Control*. 2026. arXiv: 2605.26361 [math.OC].

[15] Shengbo Wang and Nian Si. *Non-Rectangular Average-Reward Robust MDPs: Optimal Policies and Their Transient Values*. 2026. arXiv: 2603.00945 [math.OC].

[16] Shengbo Wang and Zexi Zhang. *Central Limit Theorem for Two-Time-Scale Approximate Distributionally Robust RL*. 2026. arXiv: 2605.08417 [cs.LG].

[17] Shengbo Wang, Jose Blanchet, and Peter Glynn. *Optimal Sample Complexity of Stable Discounted Markov Decision Processes*. 2023. arXiv: 2302.07477 [cs.LG].

[18] Shengbo Wang, Nian Si, Jose Blanchet, and Zhengyuan Zhou. *On the Foundation of Distributionally Robust Reinforcement Learning*. 2023. arXiv: 2311.09018 [cs.LG].

PRESENTATIONS

Q-Measure-Learning for Continuous State RL: Efficient Implementation and Convergence
Presented at IOS 2026.

Bellman Optimality of Average-Reward Robust Markov Decision Processes with a Constant Gain
Presented at INFORMS 2025, INI workshop Bridging Stochastic Control and Reinforcement Learning 2025.

Efficient Gradient Estimation for Overparameterized Stochastic Differential Equations Presentation for the 2025 APS Best Student Paper competition at INFORMS 2025.

Statistical Learning of Distributionally Robust Stochastic Control in Continuous State Spaces
Presented at INFORMS 2024.

On the Foundation of Distributionally Robust Reinforcement Learning
Presented at CISS 2024, Berkeley and Stanford Seminars.

Reinforcement Learning for Mixing Systems
Presented at INFORMS 2023.

Distributionally Robust Q-learning: Formulations, Algorithms, and Sample Complexities
Presented at SIAM OP 2023.

A Finite Sample Complexity Bound for the Distributionally Robust Q-learning
Presented at INFORMS 2022. Poster presentation at AISTATS 2023.

Distributionally Robust Q-learning: Algorithm Designs and Sample Complexities
Presented at Stanford OR Seminar.

TEACHING

Instructor at USC
ISE 220: Foundations of Probability Modeling

Course Assistant at Stanford
MS&E 220: Probabilistic Analysis
MS&E 221: Stochastic Modeling
MS&E 321: Stochastic Systems

MS&E 324 / CME 308 / MATH 228: Stochastic Methods in Engineering

Course Assistant and Tutoring at Cornell
ORIE 3510/5510: Introduction to Engineering Stochastic Processes I (Course Assistant)
SYSEN 5200: Systems Analysis Behavior and Optimization (Tutor)

PROFESSIONAL SERVICES

Journal reviewer for *Annals of Statistics*, *Journal of Machine Learning Research*, *Mathematics of Operations Research*, *Management Science*, *Manufacturing & Service Operations Management*, *Operations Research*, and *Stochastic Systems*.
Conference reviewer for Artificial Intelligence and Statistics (AISTATS), International Conference on Learning Representations (ICLR), and Neural Information Processing Systems (NeurIPS).